Bayesian Ridge & Lasso

To maximize $p(\beta|X,Y)$, posterior distribution of β.

By Bayes theorem $p(\beta|X,Y) \propto f(Y|X,\beta)\,\pi(\beta)$

↳ prior density of β

Assume $Y = X\beta + \varepsilon$, $y_i \overset{i.i.d.}{\sim} N(x_i^T\beta, \sigma^2)$ $i=1,\dots n$

Ridge Assume $\beta_j \overset{i.i.d.}{\sim} N(0,\tau^2)$, $j=1,\dots,p$

where σ^2 and τ^2 are fixed

$$p(\beta|X,Y) \propto \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(y_i - x_i^T\beta)^2}{2\sigma^2}\right\} \prod_{j=1}^{p} \frac{1}{\sqrt{2\pi\tau^2}} \exp\left\{-\frac{\beta_j^2}{2\tau^2}\right\}$$

To maximize $p(\beta|X,Y) \Leftrightarrow$ to maximize $\log(p(\beta|X,Y))$

i.e. to maximize

$$\sum_{i=1}^{n}\left\{-\frac{(y_i - x_i^T\beta)}{2\sigma^2}\right\} + \sum_{j=1}^{p}\left\{-\frac{\beta_j^2}{2\tau^2}\right\}$$

$$\Leftrightarrow \max \sum_{i=1}^{n}\left\{-\frac{(y_i - x_i^T\beta)^2}{2}\right\} + \sum_{j=1}^{p}\left\{-\frac{1}{2}\left(\frac{\sigma^2}{\tau^2}\right)\beta_j^2\right\}$$

$$\Leftrightarrow \min \sum_{i=1}^{n}(y_i - x_i^T\beta)^2 + \sum_{j=1}^{p}\left(\frac{\sigma^2}{\tau^2}\right)\beta_j^2$$

$\Leftrightarrow$ <u>Ridge optimization problem</u> with $\lambda = \frac{\sigma^2}{\tau^2}$

Lasso Assume β_j double exponential $(0, \tau)$

$$\pi(\beta_j) = \frac{1}{2\tau} \exp\left(-\frac{|\beta_j|}{\tau}\right)$$

Then $p(\beta | X, Y) \propto \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(y_i - x_i^T\beta)^2}{2\sigma^2}\right\} \prod_{j=1}^{p} \frac{1}{2\tau} \exp\left\{-\frac{|\beta_j|}{\tau}\right\}$

$\max p(\beta | X, Y) \iff \max \log p(\beta | X, Y)$

$$\iff \max \sum_{i=1}^{n} \left\{-\frac{(y_i - x_i^T\beta)^2}{2\sigma^2}\right\} + \sum_{j=1}^{p} \left\{-\frac{|\beta_j|}{\tau}\right\}$$

$$\iff \min \sum_{i=1}^{n} (y_i - x_i^T\beta)^2 + \sum_{j=1}^{p} \left(\frac{\sigma^2}{\tau}\right) |\beta_j|$$

$\iff$ lasso optimization problem with $\lambda = \frac{\sigma^2}{\tau}$